

Arcs complete outside a prescribed conic

Exact defect, equality rigidity, and $\rho_{\mathcal{C}}(16) = 9$

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Abstract

Let $\mathcal{C}$ be a nonsingular conic in $\text{PG}(2, q)$. We study arcs $A \subseteq \text{PG}(2, q) \setminus \mathcal{C}$ whose secants cover every point outside $A \cup \mathcal{C}$, and write $\rho_{\mathcal{C}}(q)$ for their minimum size. Starting from the classical first two equations for the secant-index distribution of an arc, we prove an exact defect identity for an arbitrary prescribed set of points that may remain uncovered. Its nonnegative local remainder gives exact equality and quantitative stability criteria. For a conic, writing $I_{\mathcal{C}}(A)$ for the total incidence of the secants of A with $\mathcal{C}$, it gives

$$q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor} \binom{k}{4} - \frac{I_{\mathcal{C}}(A)}{\lfloor k/2 \rfloor}$$

for every $\mathcal{C}$ -complete k -arc, together with an exact equality criterion and, for every prime power q ,

$$\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.$$

At equality, the concurrence points of disjoint secants form a matching design: they partition the edges of the Kneser graph $KG(k, 2)$ into cliques of maximum order. This gives a global rigidity theorem and a quantitative graph-theoretic form of stability. For ten points, the only conic equality case is the regular hyperoval in $\text{PG}(2, 8)$. We also prove an upper transfer by projective averaging. In the inverse direction, when $q+1 > \binom{k}{2}$ the complete uncovered locus canonically determines the parent k -arc and its projective-semilinear automorphism group. We quantify this inverse: distinct parent arcs have loci at symmetric-difference distance at least $2(q+1 - \binom{k}{2})$, and beyond this gap the locus distance bounds first the lost secants and then the lost vertices. Explicit witnesses meet the lower bound for $q = 5, 8, 9, 11$; at $q = 5$ the witness is a projective frame. Over $\mathbb{F}_{16}$, a certificate-checked exhaustive covering list shows that no nonzero quadratic contains an eight-arc's ordinary-uncovered locus while avoiding the arc; hence $\rho_{\mathcal{C}}(16) = 9$.

1 Introduction

An arc in a projective plane is a set of points no three of which are collinear. An arc is complete when it cannot be enlarged while remaining an arc; equivalently, every point outside it lies on a secant determined by two of its points. Complete arcs and the closely related problem of finding small saturating sets have a substantial literature; see, for example, Ball [5], Kim–Vu [15], and Nagy [20].

Fix a nonsingular conic $\mathcal{C} \subset \text{PG}(2, q)$. We impose the arc condition but allow the points of $\mathcal{C}$ to remain uncovered.

Definition 1.1. An arc $A \subseteq \text{PG}(2, q) \setminus \mathcal{C}$ is $\mathcal{C}$ -complete if every point of

$$\text{PG}(2, q) \setminus (A \cup \mathcal{C})$$

lies on a secant of A . Define

$$\rho_{\mathcal{C}}(q) = \min\{|A| : A \text{ is a } \mathcal{C}\text{-complete arc}\}.$$

More generally, if $\mathcal{H}$ is any set disjoint from an arc A , write

$$\mathcal{U}_{\mathcal{H}}(A) = \text{PG}(2, q) \setminus \left(A \cup \mathcal{H} \cup \bigcup_{\{a,b\} \in \binom{A}{2}} ab \right).$$

We abbreviate $U(A) = \mathcal{U}_{\emptyset}(A)$ for the ordinary-uncovered locus and retain $\mathcal{U}_{\mathcal{C}}(A)$ for the conic-relative locus. Thus A is $\mathcal{C}$ -complete exactly when $\mathcal{U}_{\mathcal{C}}(A) = \emptyset$.

The minimum is attained because a maximum-cardinality arc disjoint from $\mathcal{C}$ is maximal under insertion. Equivalently, A is maximal among arcs in the point set $\text{PG}(2, q) \setminus \mathcal{C}$. All nonsingular conics in $\text{PG}(2, q)$ are projectively equivalent, so the numerical value of $\rho_{\mathcal{C}}(q)$ does not depend on the chosen conic.

Two theorem families organize the paper. The forward theorem refines the classical secant-index equations into an exact prescribed-hole identity with a nonnegative local remainder. It yields the lower bound

$$\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}},$$

an equality design and a quantitative stability statement. Its conic specialization and a certificate-checked quadratic obstruction give $\rho_{\mathcal{C}}(16) = 9$. The inverse theorem starts from the complete uncovered locus. When $q+1 > \binom{k}{2}$, it canonically recovers the secants and then the parent k -arc; a robust form bounds lost secants and vertices from the symmetric difference of two loci. These are respectively the forward accounting and inverse reconstruction of the same secant arrangement.

The surrounding notions differ at the defining containment. Saturating sets need not be arcs, while almost-complete conic subsets select points on the conic [7]. A line hole recovers affine completeness, including the related hyperfocused constructions [21, 13, 16]; see Corollary 3.7. Korchmáros–Nagy–Szőnyi’s subgeometry-localized theorem concerns the $(q+1)$ -secants of a particular curve-derived arc [17, Theorem 7.5], rather than arbitrary holes and ordinary secants.

Complete exterior sets point in the opposite direction: their joins miss the conic [8, p. 143]. Thus

$$\text{complete exteriority} \Rightarrow \mathcal{C}(\mathbb{F}_q) \subseteq U(A), \quad \mathcal{C}\text{-completeness} \iff U(A) \subseteq \mathcal{C}(\mathbb{F}_q).$$

Equality occurs for the Clebsch witness at $q = 11$. Under the classical arc–MDS dictionary, the same locus is the set of projective distance-three syndrome directions of a codimension-three MDS system; the coding interpretation in Section 9 is secondary to the two geometric theorem families.

The point-index equations themselves are classical; compare Hirschfeld [14, Chapter 9], Ball [5], and the recent explicit complete-arc bound of Alabdullah–Hirschfeld [2]. The new step is the exact local remainder after splitting those equations over a prescribed hole and its complement.

The equality structure belongs to the established theory of matching designs. Alspach and Heinrich [4] call a family of s -edge matchings in K_n a $\text{MATCH}(n, s, \lambda)$ design when every pair of independent edges occurs in exactly λ members. Our contribution at this interface is the geometric implication: vanishing of the prescribed-hole defect canonically

produces a simple $\text{MATCH}(k, \lfloor k/2 \rfloor, 1)$ design, represented by concurrence points of secants in one projective plane. The local remainder also measures the graph edges that fail to lie in such maximum-matching cliques.

Sections 2–8 form the forward geometric route: exact accounting, lower and upper bounds, the characteristic-two refinement, and the finite obstruction. Section 9 gives the secondary coding reading and the $q = 11$ application. The independent geometric route resumes in Section 10: above a sharp incidence threshold, the complete uncovered locus determines its parent arc and automorphism group, with quantitative control under perturbation. The coding section may be skipped without affecting either geometric route.

The secant-index equations, the Lunelli–Sce $\sqrt{2q}$ scale, the arc–MDS correspondence, and the Clebsch geometry at $q = 11$ are classical inputs. We claim neither those ingredients nor the known ordinary eight-arc class count. The contributions are the prescribed-hole remainder with its equality and stability consequences, its explicit conic specialization, the uncovered-locus evaluation obstruction, and the certified relative-conic conclusions. The $q = 11$ deep-hole equality is an apparently unrecorded synthesis of classical geometry with the relative-completeness and coding dictionaries, not a claim of priority for the Clebsch configuration itself.

2 The classical secant equations

Throughout the remainder of the paper, $q \geq 3$ and $k \geq 3$. Let Π be a projective plane of order q , and let A be a k -arc in Π . Write $r_A(x)$ for the number of secants of A through $x \in \Pi \setminus A$. We usually write $r(x)$. Put

$$N = \binom{k}{2}, \quad m = \left\lfloor \frac{k}{2} \right\rfloor.$$

Lemma 2.1. *For every $x \notin A$, one has $r(x) \leq m$.*

Proof. Distinct secants through x cannot share an endpoint in A . Indeed, two lines through the same pair x, a coincide. Hence the secants through x use pairwise disjoint pairs of points of A , of which there are at most $\lfloor k/2 \rfloor$. $\square$

Proposition 2.2 (Classical first and second index equations). *For every k -arc A in a projective plane of order q ,*

$$\sum_{x \notin A} r(x) = \binom{k}{2}(q-1), \tag{1}$$

$$\sum_{x \notin A} \binom{r(x)}{2} = 3 \binom{k}{4}. \tag{2}$$

Proof. There are $\binom{k}{2}$ secants. Each contains $q+1$ points, exactly two of them in A , and therefore contributes $q-1$ incidences with points outside A . This proves (1).

For (2), count unordered pairs of distinct secants through a common external point. By Lemma 2.1, their four endpoints in A are distinct. Conversely, a four-subset of A has three partitions into two unordered pairs. Each partition gives two secants, which meet in a unique point because the plane is projective. Their intersection is outside A : otherwise one of the two secants would contain three points of the arc. Thus every four-subset contributes exactly three. $\square$

Only the projective-plane axioms and the common line size $q + 1$ enter these proofs. In particular, the second equation need not hold in a general finite linear space where two lines may be disjoint.

3 An exact defect identity for prescribed holes

The next theorem is stated for an arbitrary exceptional point set. The conic enters only in later specializations.

Let $\mathcal{H} \subseteq \Pi \setminus A$ be any set of points allowed to remain uncovered. The notation used in the identity is summarized by

$V_{\mathcal{H}}(A) = \Pi \setminus (A \cup \mathcal{H})$	required locus
$\mathcal{X}_{\mathcal{H}}(A) = \{x \in V_{\mathcal{H}}(A) : r(x) > 0\}$	covered required points
$\mathcal{U}_{\mathcal{H}}(A) = V_{\mathcal{H}}(A) \setminus \mathcal{X}_{\mathcal{H}}(A)$	uncovered required points
$I_{\mathcal{H}}(A) = \sum_{y \in \mathcal{H}} r(y)$	secant incidence spent on the holes.

The third line agrees with the uncovered-locus notation introduced in Definition 1.1; in particular, $U = \mathcal{U}_{\emptyset}$ and the conic-relative locus is $\mathcal{U}_{\mathcal{C}}$.

Splitting Proposition 2.2 over $V_{\mathcal{H}}(A)$ and $\mathcal{H}$ gives

$$\sum_{x \in V_{\mathcal{H}}(A)} r(x) = N(q-1) - I_{\mathcal{H}}(A), \quad (3)$$

$$\sum_{x \in V_{\mathcal{H}}(A)} \binom{r(x)}{2} + \sum_{y \in \mathcal{H}} \binom{r(y)}{2} = 3 \binom{k}{4}. \quad (4)$$

Theorem 3.1 (Prescribed-hole defect identity). *Let A be a k -arc in a projective plane of order q , let $\mathcal{H} \subseteq \Pi \setminus A$, and put $m = \lfloor k/2 \rfloor$. Define*

$$\Delta_{\mathcal{H}}(A) = N(q-1) - \frac{6}{m} \binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m} - |\mathcal{X}_{\mathcal{H}}(A)|.$$

It is the exact coverage capacity left after the universal second-moment and hole-incidence losses; the identity below resolves that capacity into local nonextremal-index contributions. Then

$$m\Delta_{\mathcal{H}}(A) = \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} (r(x) - 1)(m - r(x)) + \sum_{y \in \mathcal{H}} r(y)(m - r(y)). \quad (5)$$

In particular, $\Delta_{\mathcal{H}}(A) \geq 0$.

Proof. Only points of $\mathcal{X}_{\mathcal{H}}(A)$ contribute to the sum in (3). Multiplying the definition of $\Delta_{\mathcal{H}}(A)$ by m and using (3) gives

$$m\Delta_{\mathcal{H}}(A) = m \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} r(x) + (m-1)I_{\mathcal{H}}(A) - m|\mathcal{X}_{\mathcal{H}}(A)| - 6 \binom{k}{4}.$$

By (4),

$$6 \binom{k}{4} = 2 \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} \binom{r(x)}{2} + 2 \sum_{y \in \mathcal{H}} \binom{r(y)}{2}.$$

Substitution first separates the required-point and hole terms:

$$m\Delta_{\mathcal{H}}(A) = \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} \left(m(r(x) - 1) - 2 \binom{r(x)}{2} \right) + \sum_{y \in \mathcal{H}} \left((m - 1)r(y) - 2 \binom{r(y)}{2} \right).$$

The pointwise identities

$$m(r - 1) - 2 \binom{r}{2} = (r - 1)(m - r),$$

$$(m - 1)r - 2 \binom{r}{2} = r(m - r),$$

now give (5). Every term is nonnegative by Lemma 2.1. $\square$

Corollary 3.2 (Coverage and uncovered-locus bounds). *Under the hypotheses of Theorem 3.1,*

$$|\mathcal{X}_{\mathcal{H}}(A)| \leq N(q - 1) - \frac{6}{m} \binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}, \quad (6)$$

and

$$|\mathcal{U}_{\mathcal{H}}(A)| \geq |V_{\mathcal{H}}(A)| - N(q - 1) + \frac{6}{m} \binom{k}{4} + \frac{I_{\mathcal{H}}(A)}{m}. \quad (7)$$

Equality holds in either inequality if and only if

$$r(x) \in \{1, m\} \quad (x \in \mathcal{X}_{\mathcal{H}}(A)), \quad r(y) \in \{0, m\} \quad (y \in \mathcal{H}).$$

Proof. The two inequalities are respectively $\Delta_{\mathcal{H}}(A) \geq 0$ and its rearrangement using $|\mathcal{U}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| - |\mathcal{X}_{\mathcal{H}}(A)|$. The equality statement follows from the vanishing of every nonnegative term in (5). $\square$

For later use, we record the complete-outside consequence before imposing any geometry on the holes.

Corollary 3.3 (Arbitrary prescribed holes). *Let $|\mathcal{H}| = h$. If A is a k -arc disjoint from $\mathcal{H}$ and complete outside $\mathcal{H}$, then*

$$q^2 + q + 1 - k - h \leq \binom{k}{2}(q - 1) - \frac{6}{m} \binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}. \quad (8)$$

Proof. Completeness gives $|\mathcal{X}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| = q^2 + q + 1 - k - h$. Substitute this equality into (6). $\square$

The exact remainder also quantifies near equality.

Corollary 3.4 (Quantitative stability). *Assume $m \geq 3$ and put*

$$M = \{x \in \mathcal{X}_{\mathcal{H}}(A) : 2 \leq r(x) \leq m - 1\},$$

$$J = \{y \in \mathcal{H} : 1 \leq r(y) \leq m - 1\}.$$

Then

$$(m - 2)|M| + (m - 1)|J| \leq m\Delta_{\mathcal{H}}(A). \quad (9)$$

Proof. For $2 \leq r \leq m-1$, $(r-1)(m-r) \geq m-2$. For $1 \leq r \leq m-1$, $r(m-r) \geq m-1$. Apply these estimates to (5). $\square$

Thus small defect forces almost every multiply covered required point to have the maximum possible multiplicity m , while almost every hole point met by a secant also has multiplicity m . This is the precise stability content behind the equality heuristic.

3.1 Concurrency designs at equality

The equality criterion has a global incidence interpretation. Label the vertices of K_k by the points of A , so that its edges label the secants of A . Two such edges are adjacent in the Kneser graph $KG(k, 2)$ exactly when the corresponding secants have disjoint endpoint pairs.

Theorem 3.5 (Concurrency decomposition and equality rigidity). *Suppose $k \geq 4$. For each $x \notin A$ with $r(x) \geq 2$, the secants through x form an $r(x)$ -edge matching M_x in K_k , and*

$$E(KG(k, 2)) = \bigcup_{x \notin A, r(x) \geq 2} E(K_{M_x}).$$

Here K_{M_x} is the clique whose vertices are the edges of M_x .

If $\Delta_{\mathcal{H}}(A) = 0$, every M_x in this decomposition has $m = \lfloor k/2 \rfloor$ edges. Thus the family $(M_x)_{r(x)=m}$ is a simple MATCH($k, m, 1$) design. If $Z = \{x \notin A : r(x) = m\}$, then

$$|Z| = \frac{3 \binom{k}{4}}{\binom{m}{2}} = \begin{cases} (k-1)(k-3), & k \text{ even}, \\ k(k-2), & k \text{ odd}, \end{cases}$$

and every secant of A contains $k-3$ points of Z when k is even and $k-2$ points of Z when k is odd.

Proof. Secants through a point outside an arc have pairwise disjoint endpoint pairs, so they form a matching. Conversely, two secants with disjoint endpoint pairs meet in a unique point outside A . Their edge in $KG(k, 2)$ therefore belongs to exactly one concurrence clique, proving the decomposition.

If the defect vanishes, Theorem 3.1 gives $r(x) \in \{1, m\}$ off $\mathcal{H}$ and $r(x) \in \{0, m\}$ on $\mathcal{H}$. Every concurrence point has index at least two, so all the cliques above have order m . The second index equation now gives $|Z| \binom{m}{2} = 3 \binom{k}{4}$.

Fix a secant. It is disjoint from $\binom{k-2}{2}$ other secants, and each point of Z on it accounts for $m-1$ of them. Hence it contains $\binom{k-2}{2} / (m-1)$ points of Z , which has the two displayed values. $\square$

After any projective duality, the $\binom{k}{2}$ secants become points. They lie on k star lines of size $k-1$, one for each point of A , and on the $|Z|$ matching lines of size m . Two dual secant-points lie on a star line when their endpoint pairs meet and on a unique matching line when they are disjoint. Thus zero defect does not merely prescribe the indices: it gives a rank-three projective realization of the star-matching pairwise-balanced design on the edges of K_k .

The same decomposition strengthens Corollary 3.4 from a count of nonextremal centres to a count of affected pairs of secants.

Corollary 3.6 (Bad-edge stability). *Let E_{bad} be the set of edges of $KG(k, 2)$ lying in a concurrence clique of order $2 \leq r(x) \leq m-1$. Then*

$$|E_{\text{bad}}| \leq \frac{m(m-1)}{2} \Delta_{\mathcal{H}}(A).$$

Proof. At an off-hole centre of index r ,

$$\binom{r}{2} \leq \frac{m-1}{2}(r-1)(m-r),$$

and at a hole centre the defect weight $r(m-r)$ is larger. Sum these inequalities and apply (5). $\square$

The line-hole case now makes the connection with complete affine arcs explicit.

Corollary 3.7 (Complete affine arcs). *Let L_∞ be a line in a projective plane of order q , and let A be a k -arc disjoint from L_∞ . Then A is complete as an arc in the affine plane $\Pi \setminus L_\infty$ if and only if it is complete outside L_∞ . In that case*

$$q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{1}{m}\binom{k}{2}. \quad (10)$$

Equality holds if and only if every affine point outside A has secant index in $\{1, m\}$ and every point of L_∞ has secant index in $\{0, m\}$.

Proof. Maximality in the affine plane is equivalent to every affine point outside A lying on a secant of A , which is exactly completeness outside L_∞ . Every secant meets L_∞ once, so $I_{L_\infty}(A) = \binom{k}{2}$. Now apply Corollaries 3.3 and 3.2; the latter also gives the equality criterion. $\square$

4 Specialization to a conic

Let $\mathcal{C}$ be a nonsingular conic in $\text{PG}(2, q)$, and let $A \subseteq \text{PG}(2, q) \setminus \mathcal{C}$ be a k -arc. Since $|\mathcal{C}| = q+1$, there are $q^2 - k$ points outside $A \cup \mathcal{C}$. Write

$$I_{\mathcal{C}}(A) = \sum_{y \in \mathcal{C}} r(y) = \sum_{\ell \text{ an } A\text{-secant}} |\ell \cap \mathcal{C}|.$$

Corollary 4.1 (Conic-loss and uncovered-locus inequalities). *For every such arc,*

$$|\mathcal{U}_{\mathcal{C}}(A)| \geq q^2 - k - \binom{k}{2}(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{C}}(A)}{m}. \quad (11)$$

If A is $\mathcal{C}$ -complete, then

$$\boxed{q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{C}}(A)}{m}.} \quad (12)$$

Proof. Apply Corollary 3.2 with $\mathcal{H} = \mathcal{C}$, using $|\mathcal{V}_{\mathcal{C}}(A)| = q^2 - k$ and $I_{\mathcal{H}}(A) = I_{\mathcal{C}}(A)$. If A is $\mathcal{C}$ -complete, then $\mathcal{U}_{\mathcal{C}}(A) = \emptyset$, and the first inequality rearranges to (12). $\square$

The first correction in (12) accounts for unavoidable concurrence among secants; the last accounts for secant incidence spent on the exempt conic.

Corollary 4.2 (Even-size equality spectrum). *Let $k \geq 6$ be even. If a $\mathcal{C}$ -complete k -arc has zero defect, then*

$$q \in \left\{ k-2, \binom{k-1}{2}, \binom{k-1}{2} + 1 \right\}.$$

In the first two cases every point of $\mathcal{C}$ has index $k/2$; in the third, exactly $k - 1$ conic points have that index.

In particular, no zero-defect $\mathcal{C}$ -complete eight- or twelve-arc exists in a Desarguesian plane, while a zero-defect ten-arc can occur only for $q = 8$ or $q = 37$. In fact, Theorem 5.2 excludes $q = 37$ and shows that the unique ten-point equality case is the regular hyperoval $\mathcal{C} \cup \{N\}$ in $\text{PG}(2, 8)$, where N is the nucleus of $\mathcal{C}$.

Consequently, if $Q \geq 4$ is an even prime power and $d \geq 2$, no $\mathcal{C}$ -complete $(Q + 2)$ -arc in $\text{PG}(2, Q^d) \setminus \mathcal{C}$ has zero defect. Thus a hyperoval over $\mathbb{F}_Q$ retains its secant-concurrence matching design after scalar extension but cannot attain equality in (12) over the larger field.

Proof. Put $Q = k - 2$ and let $Z = \{x \notin A : r(x) = m\}$. Then $m = Q/2 + 1$ and, by Theorem 3.5, $|Z| = Q^2 - 1$. Put $s = |Z \cap \mathcal{C}|$. Every positive conic index equals m , so $I_{\mathcal{C}}(A) = ms$. Substitution in the zero-defect identity and factorization give

$$s = q + 1 - \frac{(q - Q)(2q - Q(Q + 1))}{2}.$$

The elementary arc bound $k \leq q + 2$ gives $q \geq Q$. Since $0 \leq s \leq q + 1$, either $q = Q$ or $q \geq B := Q(Q + 1)/2$. Write $q = B + t$ in the latter case. Then

$$s = q + 1 - t(q - Q).$$

For $t \geq 2$, already at $t = 2$ the subtracted term exceeds $q + 1$ by

$$B - 2Q + 1 = \frac{(Q - 1)(Q - 2)}{2} > 0,$$

and the excess increases with t . Hence $t \in \{0, 1\}$. The displayed values of s at $q = Q, B, B + 1$ are respectively $q + 1, q + 1, Q + 1$.

For $k = 8, 10, 12$, the three candidate orders are respectively

$$\{6, 21, 22\}, \quad \{8, 36, 37\}, \quad \{10, 55, 56\}.$$

Since a Desarguesian plane has prime-power order, this proves the stated small-size consequences.

For the final assertion, $Q^d \geq Q^2 > B + 1$ when $Q \geq 4$ and $d \geq 2$.

For the ten-point refinement, Theorem 3.5 produces a rank-three MATCH(10, 5, 1) design. Apply Theorem 5.2. Over $\mathbb{F}_8$, the regular hyperoval is $\mathcal{C} \cup \{N\}$ up to projectivity; its other 63 plane points are exactly the five-secant concurrence centres, so it is $\mathcal{C}$ -complete and has zero defect. $\square$

Define

$$L_1(q) = \min \left\{ k \geq 3 : q^2 - k \leq \binom{k}{2}(q - 1) \right\},$$

$$L_2(q) = \min \left\{ k \geq 3 : q^2 - k \leq \binom{k}{2}(q - 1) - \frac{6}{\lfloor k/2 \rfloor} \binom{k}{4} \right\}.$$

Then

$$\rho_{\mathcal{C}}(q) \geq L_2(q) \geq L_1(q). \quad (13)$$

Indeed, for $q \geq 3$ a set of at most two points has at most one secant, and that secant together with the set and the conic contains at most $2q + 2 < q^2 + q + 1$ points. Thus a $\mathcal{C}$ -complete

arc has at least three points, so the moment bound applies to the unrestricted minimum defining $\rho_{\mathcal{C}}(q)$. The definition of $L_2(q)$ involves only integer arithmetic after multiplication by $\lfloor k/2 \rfloor$.

For calculation, the corrected capacity has the parity-dependent forms

$$\binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor} \binom{k}{4} = \begin{cases} \frac{k-1}{2}(k(q-1) - (k-2)(k-3)), & k \text{ even,} \\ \frac{k}{2}((k-1)(q-1) - (k-2)(k-3)), & k \text{ odd.} \end{cases} \quad (14)$$

For comparison, some values are

q	5	8	9	11	13	16	17	19
$L_1(q)$	4	5	5	6	6	7	7	7
$L_2(q)$	4	6	6	6	7	8	8	8

Thus the second-moment correction alone proves, for example, that a $\mathcal{C}$ -complete arc in $\text{PG}(2, 16)$ has at least eight points.

The routine algebra behind the additive bound is isolated in the next lemma on the full interval needed to avoid any parity split.

Lemma 4.3 (Polynomial estimate). *For $s \geq 2$ and $0 \leq a \leq 2$, put*

$$P_s(a) = \frac{2a-3}{4}s^3 + \frac{a^2-7a+10}{4}s^2 + \frac{-3a^2+10a-8}{2}s + \frac{-a^3+5a^2-8a+6}{2}.$$

If $P_s(a) \geq 0$, then $a \geq 3/2 - 8/s$.

Proof. On $0 \leq a \leq 2$, the coefficients of s^2 , s , and 1 in $P_s(a)$ are at most $5/2$, 1, and 3, respectively. Since $s \geq 2$, their total contribution is at most

$$\frac{5}{2}s^2 + s + 3 \leq 4s^2.$$

Thus $P_s(a) \geq 0$ implies $0 \leq (2a-3)s^3/4 + 4s^2$, and division by s^2 gives the claim. $\square$

Theorem 4.4 (Explicit additive lower bound). *For every prime power q ,*

$$\boxed{\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.} \quad (15)$$

Consequently, as q tends to infinity through prime powers,

$$\liminf_{q \rightarrow \infty} (\rho_{\mathcal{C}}(q) - \sqrt{2q}) \geq \frac{3}{2}. \quad (16)$$

Proof. For $q = 2$ the right side of (15) is negative, so the assertion is immediate. Assume $q \geq 3$. Let k be the size of a $\mathcal{C}$ -complete arc and put $s = \sqrt{2q}$. The elementary first-moment inequality $q^2 - k \leq \binom{k}{2}(q-1)$ implies $k \geq s$ for $q \geq 2$: otherwise $k < s \leq q$, while $k^2 < 2q$ gives $\binom{k}{2} < q$, so its right side is smaller than $q(q-1)$ whereas $q^2 - k \geq q(q-1)$. If $k \geq s+2$, then (15) is immediate. Otherwise write $k = s + a$, so $0 \leq a < 2$ and $s \geq 2$.

Since $m = \lfloor k/2 \rfloor \leq k/2$,

$$\frac{6}{m} \binom{k}{4} \geq \frac{12}{k} \binom{k}{4} = \frac{(k-1)(k-2)(k-3)}{2}.$$

Dropping the nonnegative conic term in (12) therefore gives the necessary inequality

$$q^2 - k \leq \frac{k-1}{2}(k(q-1) - (k-2)(k-3)).$$

After substituting $q = s^2/2$ and $k = s + a$, the right side minus the left side is $P_s(a)$. It is nonnegative, so Lemma 4.3 gives $a \geq 3/2 - 8/s$, which is (15). Letting q tend to infinity through prime powers gives (16). $\square$

Remark 4.5 (The scale of the conic term). Every secant meets $\mathcal{C}$ in at most two points, so

$$0 \leq I_{\mathcal{C}}(A) \leq 2 \binom{k}{2}.$$

At the relevant scale $k \asymp \sqrt{q}$, the term $I_{\mathcal{C}}(A)/m$ is therefore $O(\sqrt{q})$, while the universal overlap correction $6\binom{k}{4}/m$ is $\Theta(q^{3/2})$. Consequently, improving the lower bound on $I_{\mathcal{C}}(A)$ within (12) can be useful for finely balanced finite cases, but cannot improve the additive constant $3/2$ in Theorem 4.4. Any stronger asymptotic result needs an additional mechanism, not merely a spectral estimate for $I_{\mathcal{C}}(A)$.

5 Equality classifications

The first two nontrivial orders illustrate the distinction between an abstract matching design and its rank-three projective realization.

Corollary 5.1 (Six and seven points). *(a) In a Desarguesian plane, if a six-arc has every intersection of two disjoint secants on a third secant, then the field has characteristic two, contains $\mathbb{F}_4$, and the arc is projectively equivalent to*

$$(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1), (1, \omega, \omega^2), (1, \omega^2, \omega), \quad \omega^2 + \omega + 1 = 0.$$

Conversely, this configuration has the stated concurrence property. In particular, every zero-defect six-arc in a Desarguesian plane has this form.

(b) In any projective plane, no seven-arc has zero defect with respect to any prescribed hole set.

Proof. For six points, the three intersections of opposite sides of any complete quadrangle must all lie on the secant through the remaining pair. Normalize four vertices to

$$(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1).$$

Their diagonal points are

$$(1, 1, 0), (1, 0, 1), (0, 1, 1).$$

Their determinant is -2 , so they are collinear exactly in characteristic two, on $x + y + z = 0$; that line contains the remaining pair. Write a fifth point as $(1, t, 1 + t)$. Applying the diagonal-line condition to the quadrangle formed by the first three frame points and this fifth point gives the diagonal line

$$t(1 + t)x + (1 + t)y + tz = 0.$$

It must contain the remaining frame point $(1, 1, 1)$, which gives $t^2 + t + 1 = 0$; its intersection with $x + y + z = 0$ is the sixth point $(1, t^2, t)$. Direct substitution verifies all required concurrences.

For seven points, Theorem 3.5 would produce a $\text{MATCH}(7, 3, 1)$ design. Alspach and Heinrich [4, Theorem 3.1] proved that no such design exists. $\square$

Theorem 5.2 (Rank-three ten-point classification). *Let K be a field. Among the simple $\text{MATCH}(10, 5, 1)$ designs, only the regular-hyperoval design can have a rank-three projective realization. It has one over K if and only if*

$$\text{char } K = 2, \quad \text{and} \quad \mathbb{F}_8 \subseteq K.$$

The other abstract design has no rank-three realization over any field.

Proof. Mathon's classification [19], as recorded by Alspach and Heinrich [4], gives two abstract designs. The exact enumeration in Appendix A independently constructs representatives of both classes. After four points are normalized to a projective frame, their 63 matching blocks give 189 integral concurrency equations. Exact lift identities exclude both designs in odd characteristic and exclude the second design in characteristic two. For the regular-hyperoval design, the nondegenerate basis instead forces a root of $T^3 + T + 1$, and hence $\mathbb{F}_8 \subseteq K$. The regular hyperoval over $\mathbb{F}_8$ and scalar extension prove the converse. Appendix A records the certificate domains and replay boundary. $\square$

6 An upper-bound transfer from complete arcs

Let $t_2(2, q)$ denote the smallest size of an ordinary complete arc in $\text{PG}(2, q)$.

Theorem 6.1 (Projective averaging). *Let $H \subseteq \text{PG}(2, q)$ and let B be an ordinary complete b -arc. If*

$$b|H| < q^2 + q + 1, \tag{17}$$

then some projective image of B is disjoint from H and complete outside H .

Proof. Choose a uniformly random projectivity $g \in \text{PGL}(3, q)$. Point transitivity gives

$$\mathbb{E}|gB \cap H| = \frac{b|H|}{q^2 + q + 1} < 1.$$

Since the intersection size is a nonnegative integer, it is zero for some g . Projectivities preserve ordinary completeness, so gB is complete outside H . $\square$

Corollary 6.2 (Conic transfer). *If $t_2(2, q) \leq q$, then*

$$\rho_{\mathcal{C}}(q) \leq t_2(2, q). \tag{18}$$

More generally, every complete b -arc with $b \leq q$ has a projective image disjoint from the prescribed conic.

Proof. Take $H = \mathcal{C}$, so $|H| = q + 1$. For integral $b \leq q$ one has $b(q + 1) < q^2 + q + 1$, and Theorem 6.1 applies. $\square$

Kim and Vu proved that, for all sufficiently large q , every projective plane of order q has a complete arc of size at most $\sqrt{q}(\log q)^c$ for an absolute constant c [15]. This is less than q for large q , so Corollary 6.2 gives the following immediate consequence.

Corollary 6.3. *For some absolute constant c and all sufficiently large prime powers q ,*

$$\rho_{\mathcal{C}}(q) \leq \sqrt{q}(\log q)^c.$$

The transfer does not exploit the conic and is unlikely to be sharp. It nevertheless supplies a uniform upper bound without having to preserve the arc condition during a separate random covering construction.

7 Even characteristic and the nucleus

Assume q is even and let ν be the nucleus of $\mathcal{C}$. The nucleus is outside $\mathcal{C}$, all tangents to $\mathcal{C}$ pass through ν , and every line through ν is one of these $q + 1$ tangents.

Proposition 7.1 (Case $\nu \in A$). *Let A be a k -arc disjoint from $\mathcal{C}$ with $\nu \in A$. Exactly $k - 1$ A -secants are tangent to $\mathcal{C}$. In particular,*

$$I_{\mathcal{C}}(A) \geq k - 1, \quad I_{\mathcal{C}}(A) \equiv k - 1 \pmod{2}.$$

If A is $\mathcal{C}$ -complete, then

$$q^2 - k \leq \binom{k}{2}(q - 1) - \frac{6}{m} \binom{k}{4} - \frac{k - 1}{m}. \quad (19)$$

Proof. For every $a \in A \setminus \{\nu\}$, the line νa is tangent to $\mathcal{C}$, giving $k - 1$ tangent secants. No line joining two points of $A \setminus \{\nu\}$ can be tangent: every tangent contains ν , which would place three points of A on one line. Hence these are exactly the tangent A -secants.

Every tangent contributes one to $I_{\mathcal{C}}(A)$, while every other line contributes zero or two. This proves the lower bound and parity statement. Inequality (19) follows from Corollary 4.1. $\square$

Proposition 7.2 (Case $\nu \notin A$). *Let A be a $\mathcal{C}$ -complete arc with $\nu \notin A$. Then*

$$1 \leq r(\nu) \leq m,$$

the tangent A -secants are exactly the secants through ν , and

$$I_{\mathcal{C}}(A) \equiv r(\nu) \pmod{2}, \quad I_{\mathcal{C}}(A) \geq r(\nu) \geq 1.$$

Consequently,

$$q^2 - k \leq \binom{k}{2}(q - 1) - \frac{6}{m} \binom{k}{4} - \frac{1}{m}. \quad (20)$$

Proof. The nucleus is a required point outside $A \cup \mathcal{C}$, so relative completeness gives $r(\nu) \geq 1$; Lemma 2.1 gives the upper bound. A line is tangent to $\mathcal{C}$ exactly when it passes through ν . Thus the number of tangent A -secants is $r(\nu)$. Reducing the sum of the intersection sizes $|\ell \cap \mathcal{C}|$ modulo two gives $I_{\mathcal{C}}(A) \equiv r(\nu) \pmod{2}$. The remaining claims follow from Corollary 4.1. $\square$

Corollary 7.3 (Universal even-characteristic loss). *Every $\mathcal{C}$ -complete arc in even characteristic satisfies*

$$I_{\mathcal{C}}(A) \geq 1.$$

Consequently, inequality (12) can always be strengthened by replacing its final term by at least $-1/m$.

Proof. If the nucleus belongs to A , Proposition 7.1 gives $I_{\mathcal{C}}(A) \geq k - 1 \geq 1$. Otherwise Proposition 7.2 gives $I_{\mathcal{C}}(A) \geq r(\nu) \geq 1$. $\square$

These two cases are useful structural reductions, but Remark 4.5 limits what the incidence term alone can accomplish. For example, at $(q, k) = (16, 8)$ inequality (12) would be contradicted only by $I_{\mathcal{C}}(A) > 268$, whereas the universal upper bound is $I_{\mathcal{C}}(A) \leq 56$. Thus the exact determination of $\rho_{\mathcal{C}}(16)$ below requires information beyond a lower bound on $I_{\mathcal{C}}(A)$ inserted into the present inequality.

8 Verified finite-field examples

The verified values at $q = 5, 8, 9, 11$ use the lower bound $L_2(q)$ and explicit coordinates. At $q = 5$ the witness is the projective four-frame. At $q = 16$, the same ingredients give $8 \leq \rho_C(16) \leq 9$; a checked projective classification excludes eight. The proof has three steps: the defect bound supplies the lower bound eight; a hypothetical $\mathcal{C}$ -complete eight-arc would have its ordinary-uncovered locus on a quadratic avoiding the arc; and the certificate rules out every such quadratic, while the displayed nine-point witness supplies the upper bound.

8.1 Evaluation obstruction

The algebraic rejection used in that classification is an instance of the following elementary linear-algebra observation. Linear systems of quadrics containing arcs are an established tool; see Glynn [12] and Ball [6]. Fix nonzero vector representatives of the projective points under consideration. If V is a finite-dimensional linear system of homogeneous forms, write $\text{ev}_x \in V^*$ for evaluation at the representative of x . The zero or nonzero status of this evaluation is independent of the chosen representative.

Indeed, if an arc is complete outside a conic, then its ordinary-uncovered locus lies in the zero set of the conic equation while every selected point avoids that zero set. The next lemma isolates exactly the linear-algebraic obstruction used to rule this out at $q = 16$.

Lemma 8.1 (Uncovered evaluation obstruction). *Let A and U be finite sets of projective points. There is no nonzero $f \in V$ such that $U \subseteq Z(f)$ and $A \cap Z(f) = \emptyset$ if either of the following holds:*

1. *the evaluation map $f \mapsto (\text{ev}_u(f))_{u \in U}$ is injective;*
2. *for some $a \in A$, the functional ev_a belongs to the span of $\{\text{ev}_u : u \in U\}$.*

Proof. If f vanishes on U , injectivity in the first case gives $f = 0$. In the second case, applying a spanning relation for ev_a to f gives $f(a) = 0$, contrary to $A \cap Z(f) = \emptyset$. $\square$

For the full system of degree- d forms, the evaluation functionals are the degree- d Veronese images of the points, up to nonzero scalar. Thus the first alternative says that the uncovered Veronese images span the dual coefficient space; the second supplies a selected Veronese image already in their span. In particular, for the full degree- d system it is enough to exhibit $\binom{d+2}{2}$ uncovered points whose Veronese evaluation matrix is invertible. This formulation is not specific to conics.

8.2 Small values and the classification over $\mathbb{F}_{16}$

Proposition 8.2. *For the standard conic*

$$\mathcal{C} : XZ = Y^2$$

in the indicated projective planes,

$$\rho_C(5) = 4, \quad \rho_C(8) = 6, \quad \rho_C(9) = 6, \quad \rho_C(11) = 6.$$

Proof. Equation (13) gives

$$L_2(5) = 4, \quad L_2(8) = L_2(9) = L_2(11) = 6.$$

For $q = 5$, the standard four-frame has ordinary-uncovered locus

$$V(X^2 + Y^2 + Z^2 + XY + XZ + YZ),$$

a nonsingular six-point conic. An invertible coordinate change carries this conic to $XZ = Y^2$ and the frame to the four coordinates in Appendix A; the relative certificate is checked in Lean. The appendix also lists $\mathcal{C}$ -complete arcs of size six for $q = 8, 9, 11$. Their verification is described below. $\square$

For context, the companion treatment classifies all conic-filling loci for $4 \leq k \leq 7$ [1, the conic-filling classification]. That broader classification is not used in Proposition 8.2.

The ordinary classification of eight-arcs over $\mathbb{F}_{16}$ records arc counts, stabilizers, secant-index distributions, and conic-contained classes. The theorem below adds the datum needed here: the ordinary uncovered locus either imposes six independent quadratic conditions or forces every containing quadratic to meet the arc.

The proof uses an exhaustive covering list, not pairwise inequivalent representatives. The following contract isolates the mathematical completeness step; Appendix A records the normalized point encoding, generator, certificate checker, and trust boundary. Let F denote the standard projective four-frame.

Lemma 8.3 (Augmentation certificate contract). *For $4 \leq i \leq 8$, let $\mathcal{L}_i$ be a finite list of i -arcs with $\mathcal{L}_4 = \{F\}$. Suppose that for every $4 \leq i < 8$, every $P \in \mathcal{L}_i$, and every $x \notin P$ such that $P \cup \{x\}$ is an arc, the certificate supplies a child $P' \in \mathcal{L}_{i+1}$ and an invertible linear map whose induced projectivity carries $P \cup \{x\}$ onto P' . Then every eight-arc in $\text{PG}(2, 16)$ is projectively equivalent to a member of $\mathcal{L}_8$.*

Proof. Choose four points of the eight-arc. Every four-arc is a projective frame, so a projectivity carries those points to F . Insert the remaining four points in any order. At each step the stated certificate transports the legal insertion to the next list. Induction through the four layers gives a member of $\mathcal{L}_8$ projectively equivalent to the original arc. $\square$

Theorem 8.4 (Quadratic avoidance over $\mathbb{F}_{16}$). *Let A be any eight-arc in $\text{PG}(2, 16)$, and let $U(A)$ be the points outside A that lie on no secant of A . There is no nonzero homogeneous quadratic form Q such that*

$$U(A) \subseteq Z(Q) \quad \text{and} \quad A \cap Z(Q) = \emptyset.$$

The assertion includes singular quadratic forms.

Proof. Covering list. The checked list lengths are $|\mathcal{L}_5| = 4$, $|\mathcal{L}_6| = 61$, $|\mathcal{L}_7| = 454$, and $|\mathcal{L}_8| = 2633$. Every legal insertion among all 273 projective points has a checked child and projective transporter, so Lemma 8.3 proves exhaustiveness. The final length agrees with the independent class count of Al-Seraji–Al-Ogali [3, Theorem 3.8.1], but that agreement is not used.

Leaf obstruction. For 2630 leaves, six uncovered points have an invertible quadratic evaluation matrix. For each of the other three, evaluation at an arc point lies in the span of evaluations on the uncovered locus. These are the two alternatives of Lemma 8.1. Projective transport preserves the assertion, so the checked leaf result holds for every eight-arc. $\square$

Corollary 8.5 (Exact relative value over $\mathbb{F}_{16}$). *For every nonsingular conic $\mathcal{C} \subseteq \text{PG}(2, 16)$,*

$$\rho_{\mathcal{C}}(16) = 9.$$

Proof. Equation (13) gives $L_2(16) = 8$. If a $\mathcal{C}$ -complete eight-arc existed, its ordinary-uncovered locus would lie in the zero set of the nonzero quadratic defining $\mathcal{C}$, while the arc is disjoint from that zero set. This contradicts Theorem 8.4. The nine-point witness in Appendix A gives the matching upper bound. $\square$

9 Coding interpretation and the $q = 11$ application

This section is independent of the geometric proof of $\rho_C(16) = 9$. It gives a second reading of the defect identity and then applies that dictionary to the $q = 11$ witness.

Choose nonzero column representatives of a k -arc $A \subset \text{PG}(2, q)$ and form a $3 \times k$ matrix H_A . For $k \geq 4$, the code

$$C_A = \{c \in \mathbb{F}_q^k : H_A c^\top = 0\}$$

is a $[k, k-3, 4]_q$ maximum-distance-separable (MDS) code: every three columns are independent, the columns span $\mathbb{F}_q^3$, and any four columns support a dependence. Conversely, a projective parity-check system for a code with these parameters is a k -arc. This correspondence and its coset-weight consequences are classical; see Davydov–Marcugini–Pambianco [10].

We call such a code *projectively GRS* when, after permuting and scaling columns and applying a projective coordinate change, its projective parity-check columns lie on a normal rational curve. In $\text{PG}(2, q)$ this is a nonsingular conic, by the standard normal-rational-curve/GRS correspondence [23]. The next proposition translates an uncovered point into a distance-three syndrome direction and a secant index into a minimum-leader count.

Proposition 9.1 (Syndrome form of relative completeness). *Let $s \neq 0$ have projective direction $x = [s] \in \text{PG}(2, q)$. Then:*

- (i) *the minimum weight of a word c with $H_A c^\top = s$ is one, two, or three according as $x \in A$, $x \notin A$ lies on a secant of A , or x is ordinarily uncovered;*
- (ii) *if the minimum weight is two, the number of weight-two leaders is exactly $r_A(x)$;*
- (iii) *if the minimum weight is three, the number of weight-three leaders is exactly $\binom{k}{3}$; and*
- (iv) *adjoining x as one more projective parity-check column preserves the MDS property exactly when x is ordinarily uncovered.*

Consequently, A is complete outside a prescribed set $\mathcal{H}$ exactly when all its projective distance-three syndrome directions lie in $\mathcal{H}$ (and $A \cap \mathcal{H} = \emptyset$). These directions are deep-hole directions only when covering radius three has separately been established.

Proof. A weight-one representation is a scalar multiple of a selected column. A weight-two representation uses a unique unordered column pair whose projective span is a secant through x ; independence of the pair makes its two coefficients unique.

If neither case occurs, any three columns form a basis and represent s . For each three-subset the representation is unique, and none of its coefficients can vanish, so there are exactly $\binom{k}{3}$ weight-three leaders. The extension statement is the same three-column independence test. The leader count also follows as the $d = 4$ case of the general farthest-coset formula in Davydov–Marcugini–Pambianco [10, Theorem 7.7]; the deep-hole/MDS-extension formulation is standard in the Reed–Solomon setting [18]. $\square$

Thus $r_A(x)$ is the exact number of minimum weight-two leaders of every affine syndrome on direction x . All $q - 1$ nonzero scalar multiples of a projective syndrome direction have the same distance and leader count: scalar multiplication bijects their coefficient words without changing Hamming support. This accounts for every projective-to-affine factor of $q - 1$ below.

The first moment counts leaders, the second counts collisions between distinct leader supports, and Theorem 3.1 is an exact leader-collision remainder after prescribed directions are removed. Thus Corollary 3.3 is also a length obstruction for projective codimension-three MDS codes whose distance-three syndrome directions are confined to the prescribed projective set.

9.1 The Clebsch witness over $\mathbb{F}_{11}$

The six-point witness over $\mathbb{F}_{11}$ is the classical Clebsch hexagon with icosahedral A_5 symmetry; see Edge, Dye, and Storme–Van Maldeghem for the configuration and its orbit geometry [9, 11, 22]. Dye’s edge criterion and the complete-exterior formulation of Blokhuis–Seress–Wilbrink show that its secants avoid the conic, so $\mathcal{C}(\mathbb{F}_{11}) \subseteq U(A)$ [11, discussion preceding Theorem 6, p. 281] [8, pp. 143, 146]. The verified relative completeness of the witness gives the reverse inclusion and hence $U(A) = \mathcal{C}(\mathbb{F}_{11})$. The companion manuscript [1] starts from this syndrome locus and develops the intrinsic rigidity, automorphism, decoding, and factorization theory of the Clebsch hexagon code. Here we use only the locus equality, its relative-completeness consequence, and the resulting coset counts.

The equality is stronger than complete exteriority: it identifies the entire uncovered locus. By contrast, an exterior four-arc over $\mathbb{F}_7$ has $|U(A)| = (q-2)(q-3) = 20 > 8 = |\mathcal{C}(\mathbb{F}_7)|$ [8, pp. 143, 146]. Under the coding dictionary, the Clebsch equality says that the associated code has precisely the conic as its projective distance-three syndrome locus.

Proposition 9.2 (The $q = 11$ code and coset spectrum). *Let A be the six-point witness in Appendix A, and let C_A be its parity-check code. Then:*

- (i) C_A is a projectively non-GRS $[6, 3, 4]_{11}$ MDS code of covering radius three, and its projective deep-hole syndrome locus is exactly the twelve-point conic $\mathcal{C}$;
- (ii) the affine cosets of distances 0, 1, 2, 3 number $(1, 60, 1150, 120)$, and the distance-two cosets split as $(900, 150, 100)$ according as they have one, two, or three minimum-weight leaders.

Proof. Every witness triple has nonzero determinant and the six columns span $\mathbb{F}_{11}^3$, giving the MDS parameters. The 6×6 evaluation matrix of the quadratic monomials

$$X^2, XY, XZ, Y^2, YZ, Z^2$$

is nonsingular, so no conic contains all six projective columns; by the preceding definition and the classical normal-rational-curve dictionary, the code is not projectively GRS. Direct finite arithmetic gives the secant-index spectrum

$$0 : 12, \quad 1 : 90, \quad 2 : 15, \quad 3 : 10,$$

where index zero is precisely the conic. Multiplying each nonzero projective direction by its ten affine scalars gives all nonzero affine syndromes bijectively. Proposition 9.1 identifies the determinant-zero witness pairs with weight-two leader supports; uniqueness on an independent support and scalar multiplication then give the claimed coset and leader counts.

The twelve index-zero directions yield affine syndromes of distance three. Every syndrome has distance at most three because any three parity-check columns form a basis. Hence the covering radius is three, and these twelve directions are exactly the projective deep-hole locus. $\square$

10 Uncovered-locus reconstruction

At every fixed length, the complete set of one-column MDS extensions eventually remembers the projective parity-check system itself. The inverse map has two elementary stages. Put $N = \binom{k}{2}$ and $S = \text{PG}(2, q) \setminus U(A)$. First recover the secant arrangement from the gap

between $q + 1$, the number of points on a genuine secant, and N , the largest number of points in which a nonsecant line can meet the union. Then recover the vertices from the gap between their secant multiplicity $k - 1$ and the largest possible nonvertex multiplicity $\lfloor k/2 \rfloor$:

$$U(A) \longmapsto S \longmapsto \{\ell : |\ell \cap S| > N\} \longmapsto \{x : \deg(x) = k - 1\} = A.$$

The strict first gap is the only large-field hypothesis.

Proposition 10.1 (Uncovered-locus reconstruction). *Let A and B be k -arcs in $\text{PG}(2, q)$, where $k \geq 3$ and*

$$q + 1 > \binom{k}{2}.$$

If $U(A) = U(B)$ as literal subsets of the plane, then $A = B$ as unlabelled point sets.

Proof. The complement of $U(A)$ is the union of the $\binom{k}{2}$ distinct secants of A , and likewise for B . Let ℓ be a secant of A . If no secant of B were equal to ℓ , then the secants of B would cover at most $\binom{k}{2}$ points of ℓ , one per line. This is impossible because $|\ell| = q + 1 > \binom{k}{2}$ and the two secant unions are equal. Thus every secant of A is a secant of B ; since the two sets have the same finite size, their secant-line sets coincide.

It remains to recover the vertices from this line arrangement. Each point of A lies on its $k - 1$ incident secants. A point outside A lies on at most $\lfloor k/2 \rfloor$ secants: distinct secants through it use disjoint pairs of the k vertices and hence form a matching. Since $k - 1 > \lfloor k/2 \rfloor$ for $k \geq 3$, the set A is exactly the set of points incident with $k - 1$ recovered secants, and the same description recovers B . $\square$

Corollary 10.2 (Canonical inverse and automorphisms). *Under the hypotheses of Proposition 10.1, put $S = \text{PG}(2, q) \setminus U(A)$ and $N = \binom{k}{2}$. The secants of A are exactly the lines ℓ with $|\ell \cap S| > N$, and A is exactly the set of points incident with $k - 1$ such lines. Moreover,*

$$\text{Stab}_{\text{PGL}_3(q)}(U(A)) = \text{Stab}_{\text{PGL}_3(q)}(A).$$

Proof. Every secant lies in S and has $q + 1 > N$ points. A nonsecant line meets each of the N secants in at most one point, so it contains at most N points of their union S . The vertex-recovery rule is the one proved above. Finally, every semilinear projectivity stabilizing A stabilizes $U(A)$. Conversely, if $gU(A) = U(A)$, then $U(gA) = gU(A) = U(A)$, so Proposition 10.1 gives $gA = A$. $\square$

Remark 10.3 (Sharpness of the strict threshold). The strict inequality cannot in general be weakened to $q + 1 \geq \binom{k}{2}$. For the projective four-frame A over $\mathbb{F}_5$, Proposition 8.2 gives $U(A) = \mathcal{C}(\mathbb{F}_5)$, while $q + 1 = \binom{4}{2} = 6$. The stabilizer of $\mathcal{C}$ in $\text{PGL}_3(5)$ has order $|\text{PGL}_2(5)| = 120$, whereas the setwise stabilizer of a projective frame has order $4! = 24$. The conic stabilizer therefore moves A through at least five distinct four-frames, all with the same literal uncovered locus $\mathcal{C}(\mathbb{F}_5)$.

Lemma 10.4 (Perturbing a finite line union). *Let $\mathcal{L}$ and $\mathcal{M}$ be two sets of $N < q + 1$ lines in $\text{PG}(2, q)$. Put*

$$X = \bigcup_{\ell \in \mathcal{L}} \ell, \quad Y = \bigcup_{\ell \in \mathcal{M}} \ell, \quad d_X = |X \setminus Y|, \quad d_Y = |Y \setminus X|,$$

and write $r = |\mathcal{L} \setminus \mathcal{M}| = |\mathcal{M} \setminus \mathcal{L}|$ and $\delta = q + 1 - N$. Then

$$d_X, d_Y \geq r\delta - \binom{r}{2}.$$

If $r > 0$, then $d_X, d_Y \geq \delta$. If $\delta \geq 2$, then also

$$r \binom{\delta}{2} \leq \min \left\{ \binom{d_X}{2}, \binom{d_Y}{2} \right\}.$$

Proof. For each $\ell \in \mathcal{L} \setminus \mathcal{M}$, the N lines of $\mathcal{M}$ cover at most N points of ℓ . Thus $E_\ell = \ell \setminus Y$ has at least δ points. Distinct sets E_ℓ meet in at most one point, so the first inequality follows from the first two terms of inclusion–exclusion; one such set also gives $d_X \geq \delta$ when $r > 0$.

For the last inequality, each E_ℓ contains at least $\binom{\delta}{2}$ unordered pairs of points. Pairs contributed by distinct lines are disjoint, since two points determine a unique line. All these pairs lie in $X \setminus Y$. Interchanging $\mathcal{L}$ and $\mathcal{M}$ proves the corresponding statements for d_Y . $\square$

Theorem 10.5 (Separation of uncovered loci). *Let A and B be distinct k -arcs in $\text{PG}(2, q)$, where $k \geq 3$, and put*

$$\delta = q + 1 - \binom{k}{2} \geq 1.$$

Then

$$|U(A) \triangle U(B)| \geq 2\delta.$$

Proof. Apply Lemma 10.4 to the two secant sets. If they coincide, the vertex-recovery argument in Proposition 10.1 gives $A = B$. Otherwise each one-sided difference of the two line unions has at least δ points. These are the two one-sided differences of the complementary uncovered loci. $\square$

Lemma 10.6 (Secant-degree correction). *In the setting of Theorem 10.5, let r be the number of secants of A that are not secants of B , let $e_A(x)$ be the number of these lost secants through $x \in A$, and define $e_B(y)$ analogously for the gained secants. Put*

$$d_A = |U(B) \setminus U(A)|, \quad d_B = |U(A) \setminus U(B)|,$$

and set $\phi(0) = 0$ and $\phi(j) = \binom{j-1}{2}$ for $j \geq 1$. Then

$$\begin{aligned} d_A &\geq r\delta - \binom{r}{2} + \sum_{x \in A} \phi(e_A(x)), \\ d_B &\geq r\delta - \binom{r}{2} + \sum_{y \in B} \phi(e_B(y)). \end{aligned}$$

Proof. Let Y be the union of the secants of B and, for every lost secant ℓ , put $E_\ell = \ell \setminus Y$. Then $D_A = U(B) \setminus U(A)$ is the union of the E_ℓ , each of which has at least δ points. If $n(p)$ is the number of lost secants through p , then

$$|D_A| = \sum_{\ell} |E_\ell| - \sum_{p \in D_A} (n(p) - 1).$$

Every pair of lost secants has one intersection, so

$$\binom{r}{2} = \sum_p \binom{n(p)}{2}.$$

At an arc vertex x , one has $n(x) = e_A(x)$. If $x \in D_A$, its contribution to the difference between these two overlap sums is

$$\binom{e_A(x)}{2} - (e_A(x) - 1) = \phi(e_A(x));$$

if $x \notin D_A$, it is $\binom{e_A(x)}{2} \geq \phi(e_A(x))$. Pairs meeting at distinct arc vertices are disjoint. Subtracting the remaining pair intersections gives the bound for d_A ; interchange A and B for d_B . $\square$

Proposition 10.7 (Quantitative uncovered-locus reconstruction). *Let A and B be k -arcs in $\text{PG}(2, q)$, where $k \geq 3$, and put*

$$N = \binom{k}{2}, \quad \delta = q + 1 - N \geq 2, \quad t = |U(A) \triangle U(B)|, \quad s = \lfloor t/2 \rfloor.$$

If r is the number of secants lost from A to B , then

$$r \leq \left\lfloor \frac{s(s-1)}{\delta(\delta-1)} \right\rfloor.$$

If $a = |A \setminus B|$ and $h = \lfloor (k-1)/2 \rfloor$, then

$$2r \geq a(k-1 - \min\{a, \lfloor k/2 \rfloor\}), \quad |A \triangle B| \leq 2 \left\lfloor \frac{2r}{h} \right\rfloor.$$

Proof. The smaller one-sided difference of the secant unions has size at most s . The pair bound in Lemma 10.4 gives the estimate for r .

For $x \in A \setminus B$, common secants through x form a matching on B . No matching edge can use two points of $A \cap B$, since its line would then contain three points of A . Every edge therefore uses a distinct point of $B \setminus A$, so there are at most $\min\{a, \lfloor k/2 \rfloor\}$ such secants. Each missing vertex is incident with at least $k-1 - \min\{a, \lfloor k/2 \rfloor\}$ lost secants. Summing lost degrees gives the first inequality. Replacing the minimum by $\lfloor k/2 \rfloor$ gives $ha \leq 2r$, and $|A \triangle B| = 2a$ gives the second. $\square$

Remark 10.8 (Range and line-arrangement obstruction). The proposition is nonasymptotic and field-uniform. Along any family for which $t = o(\delta\sqrt{N})$, it gives $r = o(N)$ and hence $|A \triangle B| = o(k)$. The scale of the line-union lemma cannot be improved for arbitrary line families. If two N -line families share $N - r$ lines and their $N + r$ lines have no triple concurrence, then, whenever $N + r \leq q + 1$,

$$d_X = d_Y = r\delta - \binom{r}{2}.$$

Indeed, every lost line meets the N comparison lines in distinct points, while its intersections with the other lost lines account for the $\binom{r}{2}$ repetitions. A pencil is less adverse: concurrency merges only $r - 1$ of these points. Any improvement at this scale must therefore use the secant realization, not merely the two line unions. The degree correction in Lemma 10.6 is the first such improvement: whenever the lost-edge graph has vertices of degree at least three, its forced concurrence recovers part of the Bonferroni loss.

11 Conclusion and further questions

The two main theorems read the same secant arrangement in opposite directions. The defect identity turns its coverage multiplicities into an exact local remainder, retaining equality and stability data lost by the usual scalar count. Conversely, the uncovered locus recovers the arrangement and then its vertices; distinct parents are separated by at least $2(q+1 - \binom{k}{2})$ points, with quantitative control beyond that gap. Projective averaging and the quadratic evaluation obstruction supply the principal applications, including $\rho_C(16) = 9$.

Problem 11.1. Determine whether Proposition 10.7 can be strengthened by using the rank-three secant realization together with the prescribed-hole defect. The lost-edge degree correction already detects forced vertex concurrence, but Lemma 10.4 is sharp to first order for unrestricted line arrangements; a defect-coupled bound must also go beyond combinatorial bad-edge stability.

Problem 11.2. Construct $\mathcal{C}$ -complete arcs of size $O(\sqrt{q})$, or prove that this is impossible for an infinite family. Theorem 6.1 currently supplies only $O(\sqrt{q}(\log q)^c)$.

Problem 11.3. Develop the higher-dimensional analogue for caps complete outside a prescribed quadric in $\text{PG}(n, q)$. The evaluation lemma already applies to arbitrary feature maps and Veronese degrees; the missing ingredient is an appropriate exact secant-moment remainder for caps.

A Computational and formal verification

The body separates mathematical reductions from finite exclusion. The corresponding evidence and trust boundaries are summarized below. The source supplement’s `TRUST.md` manifest records exact commands, toolchain versions, SHA-256 digests, theorem names, and frozen outputs.

- The companion Lean library checks the general theorem chain with the standard axioms `propext`, `Classical.choice`, and `Quot.sound`, and with no `sorry`, custom axiom, or `native_decide`.
- For $q = 16$, Lean checks transition coverage for lists of lengths 4/61/454/2633, the 2630+3 leaf profile, and the arbitrary-eight-arc theorem. Generator labels and deduplication are not trusted.
- The $q = 11$ code and syndrome claims are kernel-checked; the NRC/GRS dictionary is the cited classical input. Independent programs regenerate the finite data.
- Mathon’s theorem supplies abstract completeness for the ten-point classification through Alspach–Heinrich. The paper-local certificate checks both classes and all 189 concurrency equations; Singular exact arithmetic is trusted.

Lean also proves the operational asymptotic statement that every constant below $3/2$ is eventually a lower bound; its literal `liminf` wrapper assumes coboundedness. The rules-only witness files `ExampleChecks/Q5.lean` through `Q16.lean` contain the coordinates and field encodings referenced by Appendix A.

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